Week 6: Study Sheet

EE0849 Introduction to Robotics - Trajectory Generation

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1 Cubic Polynomial Trajectory

1.1 Problem 1.1

A joint must move from $q_0 = 10°$ to $q_f = 70°$ in $t_f = 3$ s with zero velocity at both endpoints. Using a cubic polynomial $q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$, find the coefficients.

1.1.1 Solution

Boundary conditions: $q(0) = 10$, $q(t_f) = 70$, $\dot{q}(0) = 0$, $\dot{q}(t_f) = 0$.

From $t = 0$: $a_0 = q_0 = 10$, $a_1 = 0$.

From the remaining two equations:

$$a_2 = \frac{3(q_f - q_0)}{t_f^2} = \frac{3(70 - 10)}{9} = 20$$

$$a_3 = \frac{-2(q_f - q_0)}{t_f^3} = \frac{-2(60)}{27} = -4.44$$

$$q(t) = 10 + 20\,t^2 - 4.44\,t^3$$

Check: $q(3) = 10 + 20(9) - 4.44(27) = 10 + 180 - 120 = 70°$ ✓

1.2 Problem 1.2

For the trajectory in Problem 1.1:

(a) What is the position at $t = 1.5$ s (midpoint)?

(b) What is the peak velocity and when does it occur?

1.2.1 Solution

(a) $q(1.5) = 10 + 20(2.25) - 4.44(3.375) = 10 + 45 - 15 = 40°$

This is exactly halfway ($40° = \frac{10+70}{2}$) — cubic trajectories with zero endpoint velocities are symmetric.

(b) Velocity: $\dot{q}(t) = 40\,t - 13.33\,t^2$

Set $\ddot{q}(t) = 0$: $40 - 26.67\,t = 0 \implies t = 1.5$ s

Peak velocity: $\dot{q}(1.5) = 40(1.5) - 13.33(2.25) = 60 - 30 = 30$ °/s

The peak occurs at the midpoint — again due to symmetry.

2 Quintic Polynomial Trajectory

2.1 Problem 2.1

Same motion as Problem 1.1 ($q_0 = 10°$, $q_f = 70°$, $t_f = 3$ s) but now also require zero acceleration at both endpoints. Find a_0 through a_5.

2.1.1 Solution

From $t = 0$: $a_0 = 10$, $a_1 = 0$, $a_2 = 0$ (since $\dot{q}_0 = 0$ and $\ddot{q}_0 = 0$).

Set up the 3×3 system for a_3, a_4, a_5:

$$\begin{bmatrix} 27 & 81 & 243 \\ 27 & 108 & 405 \\ 18 & 108 & 540 \end{bmatrix} \begin{bmatrix} a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$$

Solving: $a_3 = \frac{10 \cdot 60}{27} = 22.22$, $a_4 = \frac{-15 \cdot 60}{81} = -11.11$, $a_5 = \frac{6 \cdot 60}{243} = 1.48$

$$q(t) = 10 + 22.22\,t^3 - 11.11\,t^4 + 1.48\,t^5$$

2.2 Problem 2.2

Compare the peak velocity of the cubic (Problem 1.2) and quintic (Problem 2.1) trajectories.

2.2.1 Solution

Cubic peak velocity: 30 °/s (from Problem 1.2)

Quintic peak velocity: $\dot{q}(t) = 66.67\,t^2 - 44.44\,t^3 + 7.41\,t^4$

At midpoint $t = 1.5$: $\dot{q}(1.5) = 66.67(2.25) - 44.44(3.375) + 7.41(5.0625) = 150 - 150 + 37.5 = 37.5$ °/s

The quintic has a **higher peak velocity** (37.5 vs 30 °/s) to cover the same distance in the same time — because it accelerates more gently (zero acceleration at endpoints), it must go faster in the middle to compensate.

3 Trapezoidal Velocity Profile

3.1 Problem 3.1

A joint must move $d = 90°$ with $v_{max} = 60$ °/s and $a_{max} = 40$ °/s². Compute the trapezoidal profile parameters.

3.1.1 Solution

Acceleration time: $t_a = \frac{v_{max}}{a_{max}} = \frac{60}{40} = 1.5$ s

Acceleration distance: $d_a = \frac{1}{2} a_{max} \cdot t_a^2 = \frac{1}{2}(40)(2.25) = 45°$

Check: $2d_a = 90° = d$, so the cruise phase just barely exists with $t_c = 0$.

Actually $2d_a = 90° = d$ exactly, meaning the profile is **triangular** — the joint reaches v_{max} exactly at the halfway point and immediately decelerates. No cruise phase.

Total time: $t_f = 2t_a = 3.0$ s

3.2 Problem 3.2

Same motion ($d = 90°$) but with $v_{max} = 45$ °/s and $a_{max} = 40$ °/s². Now compute the profile.

3.2.1 Solution

Acceleration time: $t_a = \frac{45}{40} = 1.125$ s

Acceleration distance: $d_a = \frac{1}{2}(40)(1.125)^2 = 25.3°$

Check: $2d_a = 50.6° < 90°$ — cruise phase exists.

Cruise distance: $d - 2d_a = 90 - 50.6 = 39.4°$

Cruise time: $t_c = \frac{39.4}{45} = 0.876$ s

Total time: $t_f = 2(1.125) + 0.876 = 3.126$ s

3.3 Problem 3.3

For Problem 3.2, what happens if you increase v_{max} to 100 °/s (keeping $a_{max} = 40$ °/s²)?

3.3.1 Solution

$t_a = \frac{100}{40} = 2.5$ s, $d_a = \frac{1}{2}(40)(6.25) = 125°$

$2d_a = 250° > 90°$ — the joint **cannot reach** v_{max}!

Triangular profile: $t_a = \sqrt{\frac{d}{a_{max}}} = \sqrt{\frac{90}{40}} = 1.5$ s

Peak velocity reached: $v_{peak} = a_{max} \cdot t_a = 40 \times 1.5 = 60$ °/s $< v_{max}$

$t_f = 2t_a = 3.0$ s

4 SLERP (Orientation Interpolation)

4.1 Problem 4.1

Interpolate halfway ($s = 0.5$) between two quaternions:

- $\mathbf{q}_0 = (1, 0, 0, 0)$ — identity
- $\mathbf{q}_f = (\frac{\sqrt{2}}{2}, 0, 0, \frac{\sqrt{2}}{2})$ — 90° about z

4.1.1 Solution

Step 1: Dot product: $\mathbf{q}_0 \cdot \mathbf{q}_f = 1 \cdot \frac{\sqrt{2}}{2} + 0 + 0 + 0 = \frac{\sqrt{2}}{2}$

Step 2: $\Omega = \cos^{-1}\left(\frac{\sqrt{2}}{2}\right) = 45°$

Step 3: Weights at $s = 0.5$:

$$w_0 = \frac{\sin(0.5 \times 45°)}{\sin 45°} = \frac{\sin 22.5°}{\sin 45°} = \frac{0.383}{0.707} = 0.541$$

$$w_f = \frac{\sin(0.5 \times 45°)}{\sin 45°} = 0.541$$

Step 4: $\mathbf{q}(0.5) = 0.541 \cdot (1, 0, 0, 0) + 0.541 \cdot (\frac{\sqrt{2}}{2}, 0, 0, \frac{\sqrt{2}}{2})$

$$= (0.541 + 0.541 \cdot 0.707,\ 0,\ 0,\ 0.541 \cdot 0.707) = (0.924,\ 0,\ 0,\ 0.383)$$

This corresponds to a **45° rotation about z** — exactly halfway between 0° and 90°. ✓

4.2 Problem 4.2

Why can't you simply average two rotation matrices to interpolate orientation?

4.2.1 Solution

If R_1 and R_2 are valid rotation matrices ($\in SO(3)$), their average $R_{avg} = \frac{R_1+R_2}{2}$ is generally:

- **Not orthogonal**: $R_{avg}^T R_{avg} \neq I$
- **Determinant** $\neq 1$: $\det(R_{avg}) \neq 1$

The result is not a valid rotation — it may include scaling, shearing, or reflection. That's why we need quaternion SLERP or other representations that stay on the rotation manifold during interpolation.

5 Conceptual Questions

5.1 Problem 5.1

Explain the difference between **path** and **trajectory**.

5.1.1 Solution

A **path** is the geometric curve the robot follows (a sequence of positions in space — the "shape" of the motion). A **trajectory** is a path with a **time law** — it specifies not just where the robot goes, but when it is at each point along the path. The trajectory includes position, velocity, and acceleration as functions of time.

5.2 Problem 5.2

A FANUC program has these two lines:

```
J P[1] 50% FINE
L P[2] 200mm/sec CNT50
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For each line, state: (a) the interpolation space, (b) whether the robot stops at the point, (c) what happens in Cartesian space.

5.2.1 Solution

Line 1: (a) Joint space interpolation. (b) FINE = robot stops exactly at P[1]. (c) The Cartesian path is curved (not a straight line) because joints interpolate independently.

Line 2: (a) Cartesian space interpolation. (b) CNT50 = robot blends through P[2] without stopping (moderate blending). (c) The Cartesian path is a straight line approaching P[2], but the robot rounds the corner instead of reaching the exact point.

5.3 Problem 5.3

You need to move a joint 120° in 2 seconds with $v_{max} = 80$ °/s. Is a trapezoidal profile feasible? If not, what is the minimum time?

5.3.1 Solution

For a trapezoidal profile with no cruise phase limit, the fastest motion uses maximum velocity throughout. Required average velocity: $\frac{120°}{2\text{s}} = 60$ °/s $< v_{max} = 80$ °/s. So yes, it is feasible — the robot doesn't even need to reach v_{max}.

But we also need to check acceleration. Without knowing a_{max}, we can compute the minimum required acceleration for the trapezoidal profile. If we cruise at 60 °/s (no need to go faster), the profile degenerates to a triangle reaching 60 °/s at the midpoint. This requires $a_{max} \geq \frac{60}{1} = 60$ °/s². If that's within motor limits, the motion is feasible in 2 seconds.